



Deep Learning Framework For Cyber Attack Rate Prediction

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I hereby declare that this dissertation is all my own work, except as
indicated in the text:

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Date: 06-06-2021

I would like to dedicate this dissertation to my mom for her immense support and love.

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work, except as specified in the text.

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Lakini Senanayaka

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As an international student, It has been an interesting time and a great opportunity to study for the MSc in Cyber Security at the University of West London. During my period of studying, I learnt a lot of new technologies, met a vast variety of interesting people who showed me the meaning of my life and truly believed in me and encouraged me to be the best.

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Abstract

At present, “Cyber Attacks” receive a great amount of attention from the community due to its highly destructive behaviors which can cause massive damages to the networks and illegal access to the sensitive information. Attackers attack different systems due to political or monetary gain. With the vast development of the technology, cyber attacks including malware, phishing, denial of service attacks are getting more common and hard to exploit from the cyber defence systems deployed on the networks. Therefore, for the purpose of protecting the companies/systems from cyber attacks and threats, the defensive mechanism of the systems also should be updated with the evolution of new technologies. Some of the major cyber attacks in 2021 can be summarized as follows. CNA Financials, which is the biggest cyber insurance firm in the USA was attacked by ransomware and it disrupted the service for 3 continuous days. The recent cyber attack on the Florida Water System also tried to poison the water supply by increasing its chemical concentrations. Similarly, Microsoft’s Exchange Server which is used by most of the government bodies and private companies was exploited by attackers and caused a global, major effect (Meharchandani, 2021).

This paper introduces a new approach of prediction framework for cyber attack rate using deep learning algorithms called BRNN-GRU aka Bi-Directional Recurrent Neural Network with Gated Recurrent Units. This framework is optimised and fine-tuned for high accuracy predictions with minimised error rates. The prediction capability of this framework can guide the network defenders to proactively allocate their resources in a cost-effective manner to reduce the severity damages of the incoming attacks or totally stop them in the defender levels. Also, this paper directs the readers to the other potential approaches to build deep learning models with elevated accuracy rates, such as the usage of Generative Adversarial Networks. Furthermore, the BRNN-GRU framework has early warning prediction capabilities.

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Introduction

The Internet has become a necessity factor in human lives. Almost everything in our day-to-day life has an impact or involvement with the internet. The Internet helps the world for quick globalization, fast information sharing and even a company can run fully over the internet without having a dedicated office space. The Internet is immersive. It can provide thousands of different services like infrastructure as a service, platform as a service and also able to communicate with sensors in automobiles, nuclear and thermal plants. Not only that, but the internet also became more important during the COVID-19 lockdowns. Most of the people had to work remotely and students had to learn remotely. All of these opportunities open up thanks to the internet. Although cyberspace can provide thousands of different services for the community, it is also not 100% safe. The implementation of new and evolving technologies open up more opportunities for attackers to exploit cyberspace. Attackers and Intruders can steal user information, exploit large organizations over the internet and make monetary losses and even make reputation damages. Hence all the users need to have extra attention when working with the internet. Some people install anti-virus software on their personal computers to protect from malware, worms - the ways attackers use to exploit the user devices.

Even though cybersecurity is a vast topic to discuss, this paper only focuses on cybersecurity at the industrial level. Different companies protect their networks from attackers in different ways. Most companies have separate cybersecurity divisions with system administrators and ethical hackers to protect their systems. Intrusion detection systems (IDS), Network-based Intrusion detection systems, Intrusion Prevention Systems, honeypots and network telescopes also play an important role as deception technologies in cybersecurity. Intrusion Detection systems monitor the network for malicious activities or policy violations and trigger alerts for the security division. Then the system administrators can analyse the alerts and take the necessary steps to stop the propagation of the attacks or make the defenders ready to mitigate the attacks. At present, most of these manual workflows are automated with different other technologies. False-positive alarms in IDS are one of the significant issues faced by the

security teams. Not only that but also, the IDS cannot predict future attacks based on patterns, some IDS cannot identify zero-day attacks and new types of attacks. Therefore the researchers have made an immeasurable commitment to design and suggest new methodologies to detect correct attacks, predict and forecast future attacks and provide prevention methods for the future attacks. By using these approaches, the system administrators can use their limited resources effectively and protect the networks. Honeypots and Network Telescopes are popular cyber defence instruments installed in networks to observe the internet traffic of a network. Honeypots are used to attract attackers by acting as real production servers and collect information on the attacks and attack types by interacting with them (Huang et al., 2019). As the internet and cybersecurity evolve, the attackers' attack patterns also evolve. In Yang Li's (Li et al., 2019) research, he explained how attackers identify the dynamic honeypots and he introduced a distributed honeypot approach to avoid the identification of honeypots. If the attacker can identify the honeypot, they do not try to access it and then installation of that honeypot becomes useless in the network. The attackers are getting smarter and trying to attack stealthily. These results provide confirmatory evidence for the need for proper detection, prediction and forecasting systems for cyberspace with high accuracy prediction rates with fewer error values.

This paper is divided into 5 main sections. First section contains an introduction to this report, the second section explains and analyses the literature related to the current status of the prediction frameworks. It also discusses different types of deception technologies, statistical properties in cyberattack data, current research work for cyber attack rate predictions, identified problem statements, also the aims and objectives of this research. Third section is dedicated to the details of the new deep learning framework called BRNN-GRU. This section explains in depth discussion of data preprocessing, analysing, model building and evaluations. Section four includes the conclusion of the report and new paths for future work. This report ends with a reference section.

Literature Review

Deception Technologies

HoneyPots and HoneyNets

A honeypot is a decoy tool in network security that is used to lure attackers to interact with it by providing the services as in the production servers. These services are just an interface for the attackers to access, but these services are designed only to capture the attacker's sensitive data without providing the proper intended service as the production servers. By doing that the honeypots try to exhaust the attackers' resources and capture more details about the attack(J. ShimeallM. Spring, 2014). This can be a partial or full duplication of the real servers, therefore the attacker tries to attack the systems without knowing they are on a forgery server. Honeypots are considered as dynamic and real-time analytical tools(Li et al., 2019). Due to this behaviour, all the traffic coming to the honeypots are considered malicious traffic. There can be traffic caused due to misconfiguration of the ports or IP addresses that are not created by the attacker. On that account, such traffic should be separated when analysing the honeypot traffic data.

There are three types of commonly used honeypots in cybersecurity.

- Low - Interaction Honeypots

These types of honeypots are capable of collecting low-level information about attacks such as source IP addresses, attack frequencies etc. These honeypots do not provide any physical environment for the attackers to access so, this type of honeypots only captures attackers' attempts and creates security alerts.

- High - Interaction Honeypots

These types of honeypots are more interactive with the attackers. It allows the attackers to compromise the system by allowing access to the services. Therefore these honeypots can capture very sensitive information about the attacks including attackers' attack patterns, attack types, originating countries etc. These types of honeypots are expensive to install.

- Medium - Interaction Honeypots

Medium interaction honeypots can capture more details about the attacks than low interaction honeypots and less information than high interaction honeypots. These types of honeypots do not provide access to the system but they are more sensitive to collect information during attack attempts.

The data in high interaction honeypots come with privacy issues as it contains more valuable information about attackers and also about the vulnerabilities of the network and services of the company.

Honeynets are a collection of honeypots designed to offer more complex deception interactions with attackers.

Network Telescopes

These are also dynamic and real-time deception tools which passively monitor the network traffic. They are built on a portion of routed IP address space in which little or legitimate traffic exists. Normally, the traffic for the unused address space and illegitimate traffic is forwarded to the network telescopes for real-time analysis. Some networks forward the whole network traffic to the network telescopes without sending any response back to the user (Hunter et al., 2013).

The major difference between Honeypot and Telescopes are,

The honeypots are separate servers which use to lure the attackers to get the attacker's attention while network telescopes passively analyse the illegitimate traffic in the production servers without setting up separate servers to lure the attackers(Hunter et al., 2013).

Studying the honeypot data to understand attack patterns, categorise and prioritize them started a few decades ago. Different researchers used different methodologies to understand the data. Some of them used visualization techniques to understand the honeypot data. For example, visualizing the ports observed in honeypots using neural

projection techniques (HERRERO et al., 2012). Even Though there are few techniques to understand the honeypot data, the most common approach is statistical analysis. With this, the researchers can understand the statistical properties exhibited in the cyberattack data.

Honeypot and Network telescope data are massive resources for researchers to study cyber attacks. Based on these datasets, different researchers have proposed and designed approaches to understand anomaly behaviours and probing in the networks, infer distance of denial of service attacks and track internet worms. Analysing the internet of things(IoT) related botnets and exploited IoT devices are also studied in (Torabi et al., 2020) & (Safaei Pour et al., 2020). Denial of service attacks during an election is studied by (Lutscher et al., 2019). Darknet traffic stabilities are studied in (Vichaidis et al., 2018). These examples show the usage of honeypot and telescope data to secure systems in different ways. Apart from the above mentioned usages, Honeypots and Network telescopes play a major role in understanding and developing cyber attack detection, prediction and forecasting models.

When I conducted the previous related work for predicting cyberattack rates, I figured out that the development of introducing frameworks started from attack graphs - which is considered as a primary level of predicting methodology. Currently, the researchers have grown from that basic level to the machine learning level of frameworks and provide many more useful details and accurate predictions to the defence systems to mitigate and protect the networks from attacks before they exploit the systems. I have conducted a survey related to cyber-attack detection, prediction and forecasting methodologies. This survey explains the history and the development of methodologies until the usage of machine learning and deep learning (Senanayaka, 2020).

In this paper, I only focus on the major statistical frameworks which were really helpful to understand the honeypot data which I used as the resource for cyber attack data. Before deep-diving into the existing frameworks, I would like to explain more technical terms to understand the concepts clearly.

Definitions

This section describes some statistical preliminaries which will be useful in discussing the concepts in this paper.

Definitions for Statistical Approaches

Stochastic Process

A stochastic(random) process is a statistical phenomenon consisting of a group of random variables $\{X_\theta\}$ indexed by a parameter θ , where θ belongs to some index set Θ . In most cases, this Θ represents time (Zhang, 2021). Hence in the context of cyber attack data, a stochastic (random) process is a statistical phenomenon consisting of a collection of random variables ordered in time.

Poisson Distributions

If a statistical distribution is able to show the number of events that is likely to occur for a given period of time, then this distribution is called a Poisson distribution. These poisson distributions have independent events which occur with a constant rate over the given time.

Also, they exhibit memoryless properties as poisson distribution considers independent events over time. Cyber attack rates are not constant at any given time, hence honeypot cyber attack data are considered not to have a Poisson distribution.

Stationarity Process

A stationary stochastic processes ensemble the same statistics for any value of time. A time series is said to be stationary if there is no systematic change in mean/trend, variance, and if it contains no strictly periodic variations. Honeypot cyber attack data are considered to have stationary distribution.

Long Range Dependence (LRD)

A stationary time sequence which instantiates a stochastic cyber attack process, then $\{X_t : t \geq 0\}$ is said to possess LRD if its autocorrelation function [2] for $0 < \beta < 1$, where h is called lag, is slowly varying for (Fang, 2018),

$$\lim_{x \rightarrow \infty} \frac{L(ix)}{L(x)} = 1 \text{ for all } i > 0 \quad [1]$$

$$p(h) = \text{Cor}(X_t, X_{t+h}) \sim h^{-\beta} L(h), \quad h \rightarrow \infty \quad [2]$$

The degree of LRD is expressed using Hurst parameter (H) which relates to β in [2] (Zhan et al., 2013)

$$\beta = H - 2H \quad [3]$$

Autoregressive Moving Average Model (ARMA)

LRD-less model $ARMA(p, q)$ is a very popular time series model. It can be described as an autoregressive moving average process of orders p and q with,

$$Y_t = \sum_{i=0}^p \Phi_i Y_{t-i} + \varepsilon_t + \sum_{j=0}^q \Phi_j \varepsilon_{t-j} \quad [4]$$

where ε_t is an independent and identical normal random variable with mean 0 and variance

$$\sigma_{\varepsilon}^2.$$

Fractional AutoRegressive Integrated Moving Average Model (FARIMA)

$FARIMA(p, q, d)$, Fractional ARIMA model is also a popular time series model which can analyse time series data with LRD phenomenon. Same p, q variables as in ARMA model, the d parameter where $0 < d < 1/2$ and $H = d + 1/2$

A stationary process X_t is called $FARIMA(p, q, d)$ if,

$$\Phi(B)(1 - B)^d X_t = \psi(B)\varepsilon_t \quad [5]$$

For some $-1/2 < d < 1/2$, where

$$\Phi(x) = 1 - \sum_{j=0}^p \Phi_j X^j, \quad \psi(x) = 1 + \sum_{j=0}^q \psi(j) X^j \quad [6]$$

Where B is the back shift operator, $BX_t = X_{t-1}$, $B^2X_t = X_{t-2}$,

Generalized AutoRegressive Conditional Heteroskedasticity Model (GARCH)

A time series $\{X_t\}$ is called GARCH process if it follows,

$$X_t = \sigma_t \varepsilon_t \quad [7]$$

This ε_t is the standard white noise.

Standard GARCH(SGARCH) will calculate its variance using the following equation.

$$\sigma_t^2 = w + \sum_{j=1}^q \alpha_j \varepsilon_{t-j}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2 \quad [8]$$

Integrated GARCH(IGARCH) model uses,

$$\Phi(B) (1 - B) \varepsilon_t^2 = \omega + (1 - \psi(B)) v_t \quad [9]$$

where $v_t = \varepsilon_t^2 - \sigma_t^2$.

Extreme Value Phenomena

Let $X_0, \dots, X_n$ are the cyber attack rates of a target for a given hour. If $u > 0$ be the threshold. Then any attack rate in the series X_i is considered to be an extreme attack

value. The extreme values formulate a point process over a state space $\chi = (0, t_n] \times (u, \infty)$ as,

$$N(A) = \sum_{i=1}^n I((t_i, X_i) \in A) \quad [10]$$

The point process can be treated as a non-homogeneous poisson process with following intensity function,

$$\lambda(t, x) = \frac{1}{\delta} \left(1 + \xi \frac{x - \mu}{\delta} \right)_+^{-1/\xi - 1} \quad [11]$$

Where $(x)_+ = \max\{x, 0\}$, $\sigma > 0$, μ , ξ are the scale, location and shape parameters respectively (Peng et al., 2017).

Standard Autoregressive Conditional Duration (Standard ACD)

$$\Psi_i = \omega + \sum_{j=1}^p a_j \Delta t_{i-j} + \sum_{j=1}^q b_j \Psi_{i-j} \quad [12]$$

Where $\omega, a_j, b_j \geq 0$ and p and q are positive numbers indicating the order of the autoregressive terms (Peng et al., 2017).

Log Autoregressive Conditional Duration (Standard ACD)

$$\Psi_i = \omega + \sum_{j=1}^p a_j \log(\Delta t_{i-j}) + \sum_{j=1}^q b_j \log(\Psi_{i-j}) \quad [13]$$

In Log ACD the researchers set $p = q = 1$, because the higher order does not increase the prediction accuracy (Peng et al., 2017) (Bauwens Giot, 2000) (Bauwens et al., 2004).

Definitions for Deep Learning Approaches

Recurrent Neural Network (RNN)

This type of neural network passes the output from the previous layer as inputs to the next layer. These networks can handle sequential data efficiently and consist of input layers, hidden layers and output layers. Not only that but also they can preprocess input of any length. The model size is not increasing with the size of the input. Weights are shared across the time and the input to the next layer is dependent on the activation function, bias and input data. These networks suffer from vanishing gradient problems due to long-range dependencies and computation being slow (AmidiAmidi, 2021).

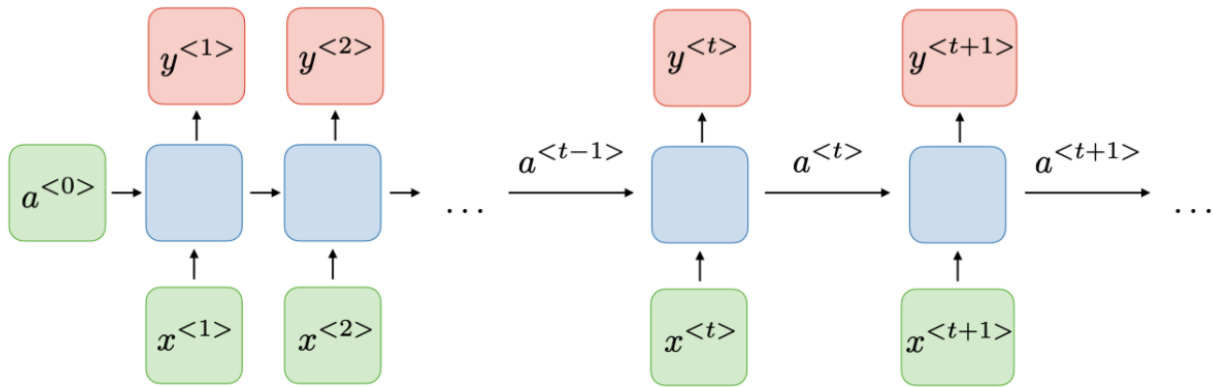


Figure 1 : Architecture of the RNN (AmidiAmidi, 2021)

For each timestamp t , the activation function can be defined as,

$$a^{<t>} = g_1(W_{aa}a^{<t-1>} + W_{ax}x^{<t>} + b_a) \quad [14]$$

Output from each layer can be defined as,

$$y^{<t>} = g_2(W_{ya}a^{<t>} + b_y) \quad [15]$$

where $W_{ax}, W_{aa}, W_{ya}, b_a, b_y$ are the coefficients that are shared temporally and g_1, g_2 are the activation functions. The following diagram illustrates a RNN cell (AmidiAmidi, 2021).

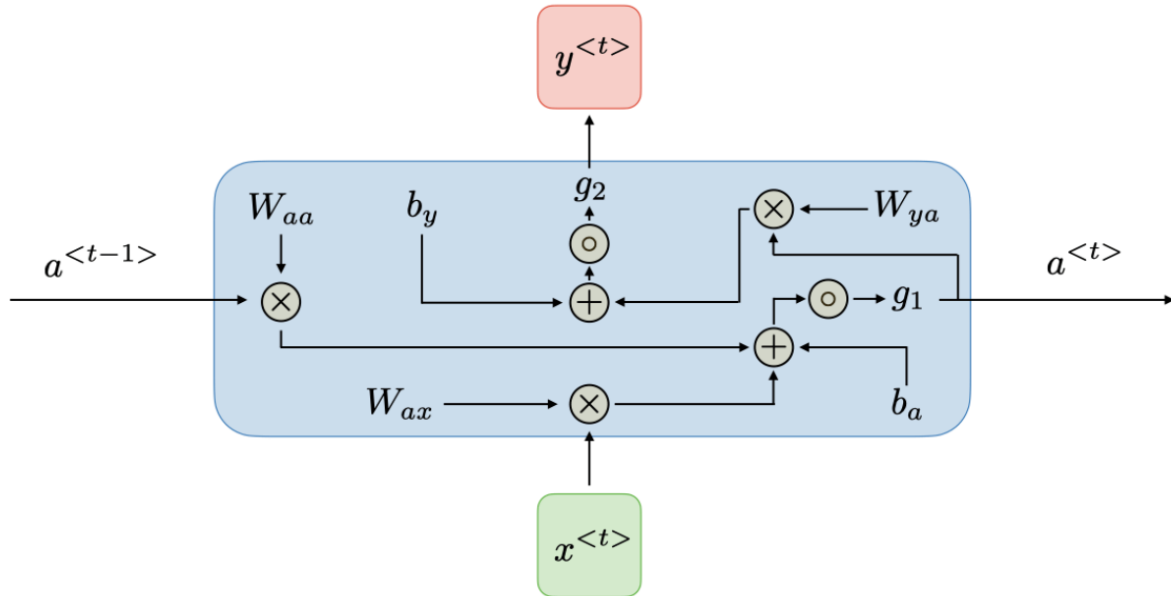


Figure 2 : Structure of an (AmidiAmidi, 2021)

Bi Directional Recurrent Neural Network (BRNN)

Unidirectional RNN learns from only the past as it only takes one sequence as the input. But bi-directional RNN learns from the past and future values. For that, it uses backpropagation and front propagation. The values from both propagations are taken as a single output of the relevant layer and input to the next layer.

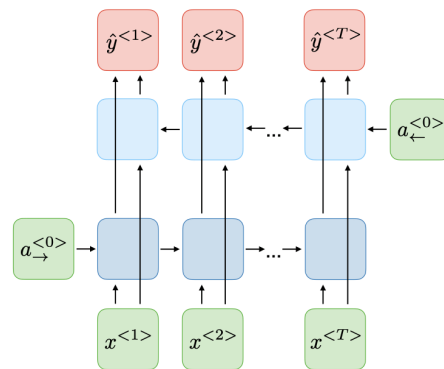


Figure 3 : Architecture of Bi-RNN (AmidiAmidi, 2021)

Long - Short Term Memory Units (LSTM)

This is another RNN structure that is mainly used to fix the vanishing gradient issue that exists on RNN. These LSTM consists of memory blocks that contain some memory cells with self connections, and store the temporal states of the network states. Also there are gates to control these memory states known as Input Gate, Forget Gate and Output Gate. Forget gate decides which information to keep and remove about the state of the previous cell (Fang, 2018).

The following figure illustrates a LSTM block with gates. In here,

i_t is the information gate, f_t is the forget gate, o_t is the output gate and c_t is the cell gate for a given time t .

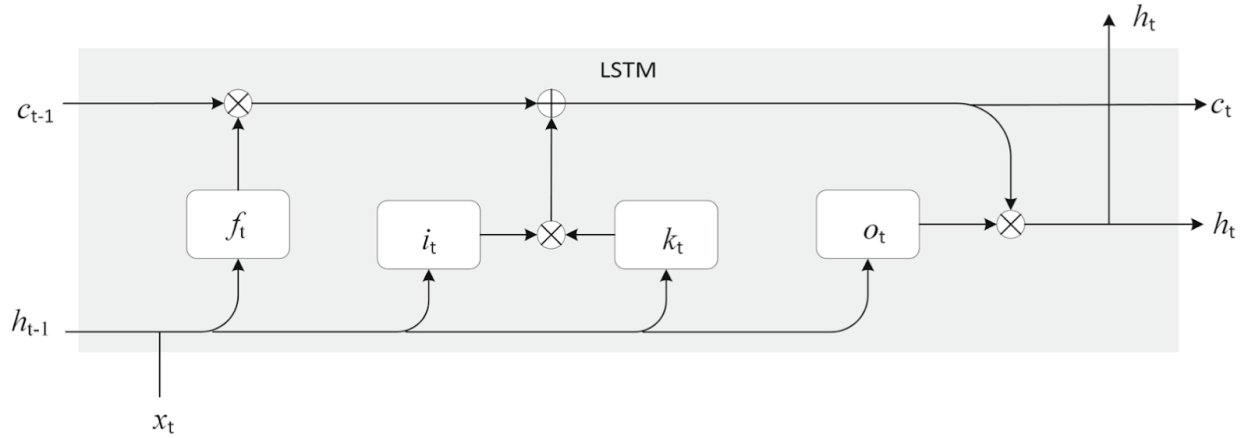


Figure 4 : Architecture of LSTM block (Fang, 2018)

Definitions for Prediction Accuracy

Let $X_m, X_{m+1}, \dots, X_z$ are the observed data for attack rate X_m for $m \leq t \leq z$ and $Y_m, Y_{m+1}, \dots, Y_z$ are the predicted results from the model. Then the prediction error can define as, $e_t = X_t - Y_t$ for $m \leq t \leq z$

Overall prediction error can be calculated using Percent Mean Absolute Deviation(PMAD),

$$PMAD = \frac{\sum_{t=m}^z |e_t|}{\sum_{t=m}^z X_t} \quad [16]$$

Overall Underestimate error (UE) can be calculated as follows.

$$PMAD' = \frac{\sum_{t=m}^z e_t}{\sum_{t=m}^z X_t} \text{ for } e_t > 0 \text{ and corresponding } X_t \quad [17]$$

UE is useful when the defender is willing to over provision some defense resources to mitigate incoming attacks.

Mean Square Error (MSE) is one of another metric to evaluate the accuracy of the predictions in a model. A model with less MSE value is considered as the best model for the predictions. MSE can be calculated as (Fang, 2018) ,

$$MSE = \sum_{i=1}^N (y_i - \hat{y}_i)^2 / N \quad [18]$$

Related Work

Cyber attack detection, prediction and forecasting using data science is not really a new topic and the history about that goes back a few decades ago. Researchers used different approaches to do detection, prediction and forecasting using discrete models such as graph theory, game theory and continuous models such as time series analysis, grey models and machine learning approaches like neural networks, SVM etc (Husak et al., 2019). The evolution of these models from attack graphs to deep neural networks are surveyed in (Senanayaka, 2020) and it includes the advantages and disadvantages of each type of approach.

As this paper mainly focused on building a prediction model for predicting cyberattack rates before a given time, the following research outcomes played a major role in

understanding the statistical features of the cyber attack time series and laid a foundation for the latest prediction models.

The statistical framework introduced by Zhenxin Zhan and his team consists of a novel approach of a framework called Gray box which is implemented based on the stochastic cyber attack process mathematical approach (Zhan et al., 2013). This model can predict attack rates 1 hour ahead of time with an accuracy of 70.2 - 82.1%. The paramount discovery of this framework is, this is the first research that found the existence of long range dependence exhibits in honeypots captured cyber attack data. Also, this model proved that the stochastic cyber-attack processes do not follow Poisson's distribution and instead can have an LRD phenomenon. Also, they identified two possible causes for existing LRD features in cyberattack data.

1. Intense Attacks (Consecutive Attacks)
2. Heavy Tailed Processes - The probability distributions of these processes are non exponentially bounded. These distributions tend to have many outliers with high values.

With keeping this in mind, this model can be instantiated on different resolutions such as network-level using IP addresses, victim level using the service types and port level using port addresses. This paper has calculated the existence of LRD in ,

- Network-level - 80%
- Victim level - 70%
- Port level - 44.5%

Therefore LDR is a must to consider when analysing and training models with honeypot cyber attack data. In Addition to that, this framework has predictive power to predict cyberattack rates ahead of time.

In technical terms, this framework consists of ARIMA and FARIMA models. In black-box models(ARIMA), the user inputs the honeypot dataset to the black box framework and

trains and gets the prediction only using a mathematical model inside the framework. It doesn't analyse the data based on the statistical properties.

Furthermore, when the statistical properties of the attacks are different, they should analyse differently to get accurate results. Hence the prediction values taken from the black box are less accurate and it is proven in this paper. To analyse the dataset with its own statistical differences, this framework provides a gray box model where the different time series analyses differently. To do that this model contains ARIMA and FARIMA models. The definitions of these models can be found in the definition section. ARIMA models are not capable of accommodating LRD features hence, this ARIMA model will be used for the attack rates which do not exhibit LRD properties. The FARIMA models are capable of handling LRD properties hence it will be considered as the LRD aware model in this framework.

The pseudocode of this framework is as follows.

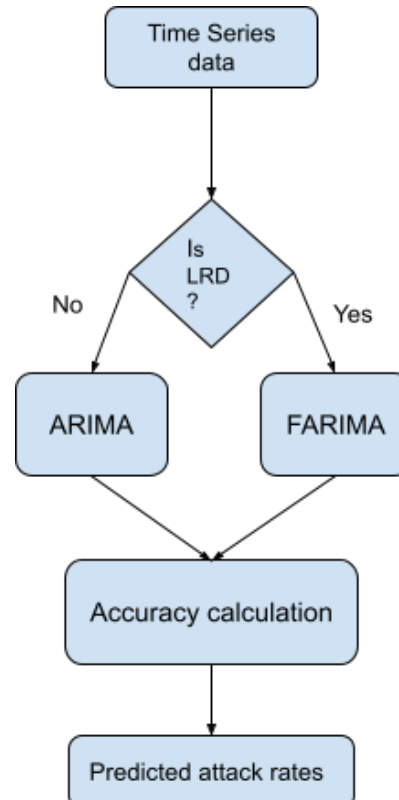


Figure 5: Pseudocode for ARMA+FARIMA model

INPUT : Observed Attack Rates sequence for the given time period $t \{X_1, \dots, X_t\}$, Number of hours ahead we need to predict the attack h

OUTPUT : Prediction Results $Y_{t+h}, Y_{t+h+1}, \dots$

PROCESS :

1. **Repeat**
2. Fit $\{X_1, \dots, X_t\}$ to obtain the finest model (M_t) from the Grey box. Time series with LRD feature will pick FARIMA and time series without LRD feature will pick ARIMA model for the prediction. Relevant p, q, d for the best fitting models will be selected using AIC criterion(CryerKung-Sik, 2008)
3. Use M_t to predict Y_{t+h} The predicted cyber attack rates at $t+h$
4. $X_{t+1} \leftarrow$ Newly observed attack rates at time $t+1$ because this framework supports for real time data
5. $t \leftarrow t+1$ Observe more data when t evolves
6. **Until** no need to predict further cyber attack rates

Algorithm 1 : Pseudocode for predicting cyber attack rates using Gray Box model(FARIMA+ARIMA)

INPUT : Observed Attack Rates sequence for the given time period $t \{X_1, \dots, X_t\}$, Number of hours ahead we need to predict the attack $h, p \in (0, 1)$ where p is the prediction starting point

OUTPUT : Prediction Accuracy

PROCESS :

1. $t \leftarrow t * p$
2. **While** $t \leq (t-h)$ **do**
3. Fit $\{X_1, \dots, X_t\}$ to obtain the finest model (M_t) from the Grey box. Time series with LRD feature will pick FARIMA and time series without LRD feature will pick ARIMA model for the prediction. Relevant p, q, d for the best fitting models will be selected using AIC criterion(CryerKung-Sik, 2008)
4. Use M_t to predict Y_{t+h} . The predicted cyber attack rates at $t+h$
5. Computer prediction error $e_{t+h} = X_{t+h} - Y_{t+h}$ In here Y_{t+h} calculated from the above algorithm
6. $t \leftarrow t+1$ Observe more data when t evolves
7. **end while**

8. Compute **PMAD, PMAD', OA, UA**
9. **return PMAD, PMAD', OA, UA**

Algorithm 2 : Pseudocode for calculating accuracy of predictions using Gray Box model(FARIMA+ARIMA)

The dataset used in this research is from UCSD Network Telescope instrumentation which captures hourly files of raw IPv4 packets of unsolicited traffic. This traffic contains a wide range of events including misconfiguration, scanning of address space by attackers or malware looking for vulnerable targets, backscatter from randomly spoofed denial-of-service attacks, and the traffic from automated spread of malware(CAIDA, 2021).

The data files in CAIDA network telescopes come as pcap files. But in this paper, the researchers have reassembled the pcap files as flows using COTS devices. These devices are capable of extracting flows from UDP and TCP traffic. In the preprocessing of data, they have disregarded the attacks against non production ports because such connections are often dropped because with non production ports, attackers cannot access the production servers. Also, flows without FIN, RST flags are also dropped if the flow timeout exceeds 60s and flow lifetime exceeds 300s.

The accuracy of the predictions is calculated using PMAD(Percent Mean Absolute Deviation). Also, this framework calculated overall accuracy(1-PMAD) and underestimation accuracy(1-PMAD') on datasets in 5 periods in order to prove the accuracy of the gray model against black box model. Appendix 1 illustrates a table containing PMAD values for ARIMA and FARIMA approaches.

The major issues I identified in this model are as follows,

- One dataset which exhibits the LRD phenomenon gave fewer accurate results for the FARIMA model. This indicates that there can be many more statistical phenomena in telescope cyber attack data

- Traditional time series model are much expensive and take more computational power
- Analysing long term forecasts and heavy traileed processes give poor performances and inaccurate results with ARIMA models (Zhai, 2005)
- ARIMA models tend to be unstable, both with respect to changes in observations and model specification.

The same team conducted another research in 2015 to find out more statistical features exhibited in the honeypot and telescope cyberattack data (Zhan et al., 2015). They found that extreme value phenomena also exist on cyber attack data. According to that, they have upgraded the above Gray box model to accommodate extreme values. In the new framework, they integrated two complementary statistical approaches called Extreme value theory (EVT) and time series theory (TST) to predict cyber attack rates in an efficient way.

According to the model, EVT can offer long term predictions 24h ahead of time and gray box TST model can predict attack rates 1h ahead of time with an accuracy of 86.0-87.9%. To accommodate extreme values in the TST model, they have introduced the FARIMA + GARCH time series approach where GARCH can accommodate extreme value phenomena and FARIMA can accommodate LRD features. The defender can use EVT predictions while making the adjustments to the resources perfectly based on the TST model prediction which will get 1h before. In this model, different GARCH models such as SGARCH, IGARCH and skewed Student-T distribution (SSTD), skewed Generalized Error Distribution (SGED) are used to facilitate different time series data with different noises.

In EVT based analysis, the team has introduced 4 models as follows,

1. M1 : The standard GPD (Generalized Pareto Distribution)
2. M2 : GPD with time-invariant shape parameter ξ but time dependent scale parameter

$$\sigma(t) = \exp(\beta_0 + \beta_1 \log(t))$$

3. M3: GDP with time-invariant scale parameter σ but time dependent shape parameter $\xi(t) = \Upsilon_1 + \Upsilon_2 t$
4. M4 : GDP with time-dependent parameters $\sigma(t) = \exp(\beta_0 + \beta_1 \log(t))$ and $\xi(t) = \Upsilon_1 + \Upsilon_2 t$

The data with stationary extreme attack rates will use M1 and if M1 cannot fit well, then the model will use non-stationary models M2,..., M4 to fit the rest of the non stationary extreme attack values. Some standard goodness-of-fit statistics and QQ plots will be used for evaluation of fitting the data to these models.

INPUT : Cyber Attack time series

OUTPUT : M1 fitting results

PROCESS :

1. Initialize quantileSet. Quantile set is an ordered set of quantiles where the maximum quantile is chosen by assuring is has at least or more than 30 extreme values
2. **For** $q \in \text{quantileSet}$ (minimum to maximum) **do**
3. Use the standard GPD to fit the extreme attack rates that are greater than the threshold quantile q
4. Evaluate good-of-fit statistics CM, AD(ChoulakianStephens, 2001) and QQPlot
5. **If** fitting is good **then**
6. Estimate GPD parameters (ξ, σ) , with extremal index θ
7. **Return** (q, ξ, σ, θ)
8. **End if**
9. **End for**
10. **Return** -1 when there is no stationary extreme value fitting

Algorithm 3 : Pseudocode for fitting stationary data to M1 in EVT model in updated Gray Box model(FARIMA+GARCH+EVT)

INPUT : Attack rate time series which cannot fit M1 model

OUTPUT : fitting results

PROCESS :

1. Initialize quantileSet.
2. **For** $q \in \text{quantileSet}$ (minimum to maximum) **do**
3. Use models M2, M3 and M4 to fit the extreme attack rates that are greater than the threshold quantile q
4. Evaluate good-of-fit statistics via AIC (Akaike Information Criterion) and QQPlot
5. **If** any of the three models fits well **then**
6. Choose the model with the minimum AIC value or choose the simplest model whose AIC value is closer to the minimum AIC value
11. **Return** ($q, \text{AIC value}$) for the selected model of M2, M3, M4
12. **End if**
13. **End for**
14. **Return** -1 when there is no extreme value fitting

Algorithm 4 : Pseudocode for fitting stationary data to M2, M3, M4 in EVT model in updated Gray Box model(FARIMA+GARCH+EVT)

The EVT model does not need a training phase like other machine learning processes do. It predicts the next attack rates based on the prediction done with the quantile sets.

The TST approach uses the following FARIMA+GARCH model.

$$\Phi(B) (1 - B)^d (y_t - \mu_t) = \psi(B) \epsilon_t \quad [19]$$

In here as well there are few models to use with different noises in the time series data.

1. M5 : FARIMA + SGARCH + SSTD
2. M6 : FARIMA + SGARCH + SGED
3. M7 : FARIMA + IGARCH + SSTD
4. M8 : FARIMA + IGARCH + SGED

The pseudocode for the TST model is as follows.

INPUT : Observed Attack Rates sequence for the given time period $t \{X_1, \dots, X_t\}$, FARIMA + GARCH family (M5,M6,M7,M8), Number of hours ahead we need to predict the attack h , lag value where $0 < l < 1$

OUTPUT : Best prediction model $M \in \{M4, \dots, M8\}$

PROCESS :

1. **For** $i = 1$ to 4 **do**
 2. $m = \lfloor nl \rfloor, j = 0$
 3. **While** $m + h \leq n$ **do**
 4. use $\{X_1, \dots, X_m\}$ to obtain the best model(M_i) from the Gray box. Gray box contains the FARIMA and GARCH models which together support LRD features and extreme value theory.
 5. Use M_i to predict attack rates $\{X_{m+1}, \dots, X_{m+h}\}$ The predicted cyber attack rates at $t + h$
 6. $m = m + h$
 7. **end while**
 8. evaluate PMAD values and AIC values for the predictions
 9. **end for**
 10. **return** $M \in \{M5, M6, M7, M8\}$ with the smallest PMAD value.
-

Algorithm 5 : Pseudocode for predicting cyber attack rates updated Gray Box model(FARIMA+GARCH+EVT)

Period	FARIMA + GARCH			FARIMA	
	model	PMAD	AIC	PMAD	AIC
I	M'_1	0.170	11.0	0.192	11.8
II	M'_1	0.185	11.4	0.195	11.9
III	M'_3	0.196	10.7	0.239	11.3
IV	M'_3	0.363	9.6	0.439	10.7
V	M'_4	0.441	11.8	0.482	12.0

Table 1 : TST based fitting of attack rates per hour using updated Gray Box model

According to the above PMAD comparisons, FARIMA+GARCH models provide highly accurate prediction results than FARIMA models.

Also the researchers have compared this new gray box model with Hidden markov model and Symbolic dynamic models which are other approaches to predict the attack

rates. The analysis proved that the FARIMA+GARCH model gives more accurate results than Hidden Markov model and Symbolic dynamics models with 1h ahead of time prediction results.

Finally the researchers calculated prediction results using both EVT and TST approaches and proved that the both approaches together can provide accurate results in 24h prior to the attack rates and more precise attack predictions in 1 h ahead of time. The following table includes all the details about their findings.

Per.	H1-24	H25-48	H49-72	H73-96	H97-120	<i>p</i> or PMAD
I	76056	76824	77088	78333	82275	0.07
	53197	60203	57370	62868	82576	
	50656	53744	52427	58183	71719	0.13
II	60910	60668	63572	62750	60752	0.17
	83157	101937	45186	38218	60274	
	72457	83073	43942	37267	51853	0.18
III	33263	32916	32836	32733	32654	0.07
	32993	30194	21476	92379	29722	
	30869	28505	21382	77747	27921	0.21
IV	29747	29622	28622	30048	30514	0.01
	23224	11671	6634	6263	8225	
	20329	9443	6674	5341	5605	0.40
V	40129	41265	42360	43526	45996	0.00
	101971	105171	114462	120221	109216	
	40341	43581	50053	54644	55007	0.41

Table 2 : Comparison between EVT and TST predictions. In here $Ha - b$ value means the predictions correspond to the time intervals between the a^{th} and the b^{th} hour. Each period has three rows. The 1st row indicates the prediction values from EVT based prediction and corresponding PMAD value in the 7th column. The second row indicates the observed maximum attack rates and third row indicates maximum attack rates predicted from the TST model with $h = 1$ and the corresponding PMAD value.

The major issues in this model are,

- This model only considers LDR and extreme value phenomena in the cyber attack rates. But there can be many more statistical features like cyclic trends and seasonal patterns in the data which need to be considered for more accurate results.
- The time complexity of the model is expressive as there are lots of loops in the algorithm and when it comes to big data, the model will take a considerable amount of time to process.

- Space complexity is also high as the model keeps another set of models to support all GARCH models with different noises and EVT models for stationary and non stationary data.
- According to table 3, the return levels of EVT based predictions of return level are often higher than the observed maximum attack rates. But TST based predictions of the maximum attack rates are often less than the observed maximum attack rates.

With the influence of the above research, Peng and his research team has developed a framework for predicting extreme cyber attack rates using only an EVT approach(Peng et al., 2017). They introduced a marked point process approach framework to fit and predict cyber attack rates. Not only that but also they introduced the existence of correlations and interdependencies between extreme values which is an important scenario to consider when predicting cyber attack rates. They found some drawbacks of the new Gray model approach introduced by (Zhan et al., 2015) as follows.

- The new Gray box method uses the classic POT(Point Over Threshold) method to model the magnitudes of the exceedances without considering the dependencies between inter exceedance times(time intervals between extreme values). Hence the accuracy of the model is low.
- Classical EVT considers the distribution of extreme values in the cyber attack data as a poisson distribution although it is not.
- The GARCH model used in the Gray model is not theoretically proved to support clustering behaviours of extreme values. Furthermore, EVT methods use clustering the extreme values via quantiles.

This framework is able to simultaneously accommodate the magnitudes of the extreme values, inter exceedance time between extreme values and the dependencies between the arrival times of extreme values.

In order to incorporate dependencies between inter exceedance with the classical POT method, the researchers proposed a marked point process approach to facilitate the

modeling of magnitudes of the extreme values using the POT method and predicting the arrival of extreme attack rates using Autocorrelation Conditional Duration (ACD) and log ACD. ACDs effectively accommodate the slow decay of autocorrelation and burst of extreme value clusters which GARCH models do not support theoretically. Also, they can dynamically adjust quantile levels in order to predict extreme cyber attack rates accurately. This ACD model consists of ground process and marked point distribution.

Value At Risk(VAR) has been used to measure cyber risk of intensive attacks. It can be described as the probability of the severity of the extreme cyber attack rates and according to these values the network defenders can allocate adequate resources to mitigate the cyber attack.

The following diagram illustrates the flow of the Marked Point process framework.

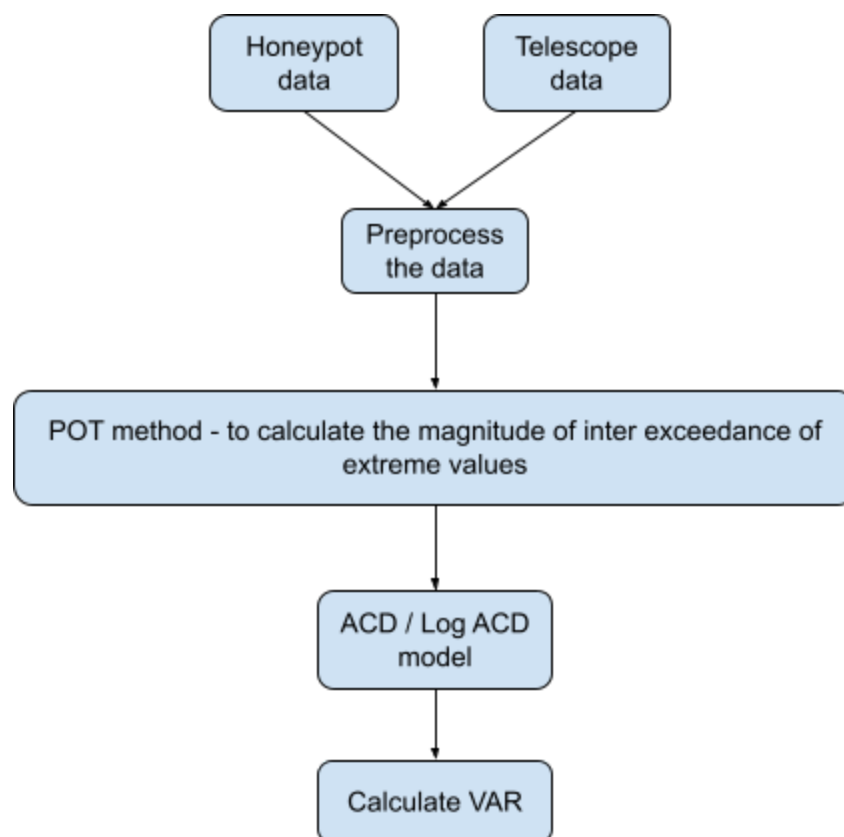


Figure 6 : Flow of the Marked Point process framework

Furthermore, this research used Honeypot data and Telescope data captured from the CAIDA network (CAIDA, 2021). As in the previous frameworks, this also preprocessed the telescope data according to (Claffy et al., 1995) and converted time series data to flows and analysed them. They followed (Almotairi et al., 2008) for the preprocessing of Honeypot data which I am going to use to introduce a deep learning framework for cyber attack rate prediction in this paper.

The major drawbacks of this model are,

- This only predicts extreme cyber attack rates prior to 1h, 4h, 10h ahead of time
- There can be more statistical features exhibits in cyber attack data such as noise, trend and seasonality features and this model does not support them

A deep learning approach is implemented by (Fang, 2018) to predict cyberattack rates called the BRNN-LSTM framework which is capable of accommodating the statistical features which we discussed in the above literature review. This model is composed of an input layer, 2 hidden layers and 1 output layer. The 2 hidden layers are the LSTM layers which are used for forwarding and backward propagation. The states of these 2 hidden layers are independent of each other and the output layers use the combined version of the last state of the 2 hidden layers as an input to do the predictions. LSTM cells help to accommodate long range dependence on the time series data. Therefore this model is capable of handling vanishing gradient issues in the RNN network. Furthermore, The training of the model follows the following objective function.

$$J = \frac{1}{2m} \cdot \sum_{i=1}^m (\hat{y}_i - y_i)^2 + \frac{\lambda}{2} (\|W\|_2^2 + \|U\|_2^2) \quad [20]$$

Where m is the size of the input, $\hat{y}_i$ and y_i are the output values from the LSTM layers and the observed values at step i, W and U are weight matrices. Also λ is the user defined penalty parameter to minimize overfitting issues.

There are two main algorithms used in this model to facilitate extreme attack rates and normal attack rates. Extreme attack rates are identified by fitting the values to another algorithm.

INPUT : Historical cyber attack time series data $\{(t, y_t) \mid t = 1, \dots, m\}$, iteration $b = 10000$, penalty parameter $\lambda = 0.001$

OUTPUT : Fitted values for various combinations of (r, l) 's

PROCESS :

```

1. for  $r \in \{20, 30, 40\}$  do
2.     Divide the data set into mini batches of size  $r$ 
3.     for  $l \in \{2, 3, 4, 5\}$  do
4.         Randomly initialize a  $l$ -layer BRNN-LSTM with parameter saved in  $\Theta$ 
5.          $j \leftarrow 0$ 
6.         while  $j \leq b$  do
7.             Computer  $j$  according to the optimization equation by performing
             forward propagation
8.             Update  $\Theta$  using the Adam optimizer
9.              $j \leftarrow j + 1$ 
10.        end while
11.        for each data point at  $t$  do
12.            Compute  $\hat{y}_t$  by performing forward propagation
13.            Fitted value  $\leftarrow \hat{y}_t$ 
14.        end for
15.        return Fitted values for combination  $(r, l)$ 
16.    end for
17. end for

```

Algorithm 6 : Pseudocode for computing fitted values in BRNN-LSTM framework (Fang, 2018)

The following algorithm is used to make mini batches to pass to the training model in order to increase the computing efficiency. After the completion of training, the model makes predictions for the testing dataset and provides prediction results and values for evaluation matrices as the output. Apart from the number of neurons per layer, the other values are hard coded in the algorithm.

INPUT : Historical time series data with in-sample set $\{(t, y_t) \mid t = 1, \dots, m\}$ and out-of-sample set $\{(t, y_t) \mid s = m + 1, \dots, n\}$ iteration $b = 10000$, penalty parameter $\lambda = 0.001$, (r, l) calculated from the Algorithm 6.

OUTPUT : Predicted values

PROCESS :

1. Split the in-sample set into mini-batches of size r
2. Randomly initialize a l -layer BRNN-LSTM, with all the parameters saved in Θ
3. $j \leftarrow 0$
4. **while** $j < b$ iteration **do**
5. **for** each mini-batch from the in-sample set **do**
6. Compute J by performing forward propagation
7. Update Θ using the Adam optimizer
8. **end for**
9. $j \leftarrow j + 1$
10. **end while**
11. Predictions $\leftarrow null$
12. **for** each data point, s , in the out-of-sample set **do**
13. Compute $\hat{y}_s$ by performing forward propagation
14. Predictions $\leftarrow \hat{y}_s$
15. **end for**
16. **return** Predicted value

Algorithm 7 : Pseudocode for predicting cyber attack rates in BRNN-LSTM framework (Fang, 2018)

This model is compared against ARIMA, ARIMA+GARCH and a hybrid model and it is evaluated against MSE, MAD(Mean Absolute Deviation), PMAD and MAPE(Mean Absolute Percentage Error) accuracy matrices. The results proved that the BRNN-LSTM model provides high accuracy results with a minimal error rate.

The major issues in this model are,

- The algorithms are complex and contain several for loops. When the dataset is too big, this will use a considerable amount of computational power and time for training and prediction
- The algorithm can tune more using hyperparameter tuning

The research problem and question

According to the gap analysis conducted for this field, different researchers contributed different ways to predict cyberattack rates with high accuracy. Most of the latest research is based on the previous ones and has fixed most of the drawbacks and gaps available on those research models. Not only that by taking all the challenges and the gap analysis conducted in the literature review it shows that current prediction methodologies need to be updated with high accuracy prediction models to warn the network defenders about cyber attacks in a few hours ahead. It will allow the defenders to allocate adequate defence resources purposefully to manage the attacks and protect the whole network(Peng et al., 2016). Not only that but also most of the statistical features exhibited in the cyber attack time series data are not identified yet. Therefore, developing a model to predict the next cyber attack with 100% accuracy is a tedious task.

The major gaps available in this research domain are,

1. Most of the currently available models use traditional statistical approaches like ARIMA, FARIMA models to predict the attack rates. These models are outdated and contain lots of drawbacks and inefficiencies. Therefore the models need to be upgraded with the latest statistical models which overcome the drawbacks and inefficiencies.
2. These models need to evolve with the technology in order to handle new attack patterns and attackers approaches.
3. The models need to be upgraded according to the newest features found in cyber attack time series data. The majority of the current models also cannot handle extreme cyberattack rates and dependencies among the features of cyberattacks.
4. Most of the currently available models are not able to self learn and tune the weights and bias according to the training cycles.
5. Most of the currently available models cannot predict the attacks more than 5-10h ahead. Also, there is no predefined value(hours) for the prediction of the attack prior to the real attack. Sometimes the longer prediction hours can

overprovision the defending resources for a particular attack, which is an expensive task.

6. Most of the available models can predict cyberattacks based on honeypot or telescope data only. They cannot predict cyber attack data based on both types of datasets.
7. Feature selection should be conducted for machine learning models
8. Most of the models are tested only for the low interaction honeypot datasets.

Not only the above mentioned problems, but we also found that the difficulty of working with statistical approaches for a person without proper knowledge of statistics. Hence we want to minimize the statistical approach model development in this paper and redirect the user to use machine learning approaches to develop models to predict cyber attack rates. The main issue found with the machine learning approaches are, the neural networks can have only a few layers to train the model and the proper features that contribute to the cyber attack predictions need to be inputted into the network. As the cyber attack patterns change every day, using the same set of features is not a good solution. Also running feature selection every time is a costly task. As a solution, deep neural networks can be used to generate models with multiple neural layers. Also the model trains by itself, so the user does not need to input the features into the model. Aside from that, deep learning models can accommodate statistical features as well. Deep learning models are easy to build and train with less computational power and less code base.

Aims and Objectives

This proposed research is planned to implement based on the research carried out by Fang (Fang, 2018) to build a deep neural network approach to predict the cyberattack rates. This proposed solution will contribute to the following gaps in the current domain.

- Increase the accuracy level of the deep neural network by increasing the performance of the network to identify long-range and short-range of dependence and high nonlinearity of the data.
- Fine tune the model by introducing efficient hyperparameters to get high accuracy rates
- Introducing new approaches in deep learning to predict cyber attack rates

The proposed new framework

Scenario

As discussed in the introduction section, predicting cyberattack rates prior to the actual attack will help the defenders to efficiently protect the company networks or mitigate the risks which could happen due to unexpected attacks. Allocating resources for a defender task is considered as a costly task and over provisioning resources for a particular attack will cause problems when allocating resources to handle other attacks. For example, if the network defenders allocated 90% of its resources (Over Provisioning of resources) for an email attack which cannot create severe damage to the system and at the same time allocated only 10% resources (under provisioning of resources) for a high severity attack, then eventually the network is unable to protect itself from all the attacks and malicious activities efficiently. Not only that but also, apart from highly secured companies, most of the other companies do not allocate more resources for network security as the defenders are expensive and they pay much more attention only to the production related resources like servers and databases.

Therefore allocating defender resources efficiently is a high priority task in order to safeguard the networks from cyber threats. There are different frameworks designed by

different researchers in order to predict the cyber attacks ahead of a given time prior to the attack. Accuracy of the predictions is the most crucial metric when designing a prediction model. The models discussed in the literature review section explains the development of the prediction models from statistical approach to deep learning approach. The researchers are trying to develop hybrid models and new deep learning models like timeGAN(Yoon et al., 2019) to work with time series data. But cyber attack rates are not just a time series data and there are only a few important features identified in cyber attack data in the current research world. Hence adapting to new deep learning models will take more research time.

Despite that fact, this paper tries to improve the accuracy of the current deep learning models - Bidirectional Recurrent Neural Network with LSTM by introducing Gated Recurrent Unit (GRU) cells and tuning the model with many more hyperparameters in order to increase the accuracy.

Definitions

Gated Recurrent Unit(GRU)

GRUs are the newest popular variant of RNN just like LSTM which can handle long term memory. RNN uses GRUs in order to handle it's gradient explosion and gradient disappearance problems that occur when it works with a large set of data (Yang et al., 2020).

GRU only consists of 2 gates as an Update Gate and a Reset Gate while LSTM consists of 3 gates called Update gate, Reset Gate and Forget Gate. The following diagram illustrates a simple GRU cell.

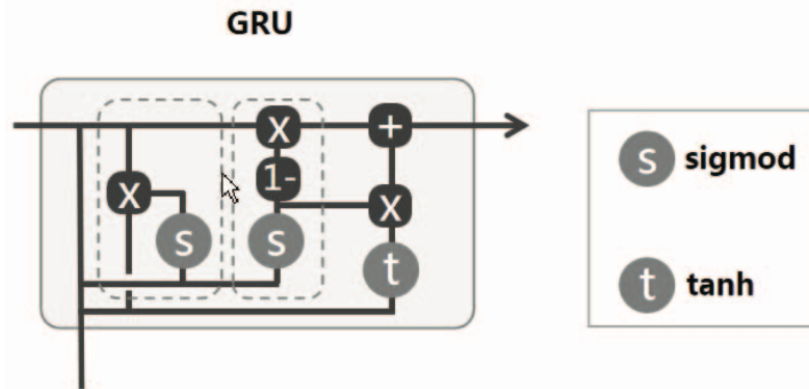


Figure 7 : Internal structure of GRU(Yang et al., 2020)

Research has been conducted by (Yang et al., 2020) to compare the performance of LSTM and GRU on a text based prediction.

Dataset Size	Accuracy Rate	Recall Rate	F1 Value	AUC
Large dataset	91.12%	97.07%	96.04%	86.98%
Small dataset	99.71%	100.67%	100.61%	99.46%

Table 3 : Average performance of GRU(Yang et al., 2020)

According to the above research conclusion, the GRU is not suitable for analysing a very large dataset with many dependency features although it uses less computational power and provides results in less time compared to LSTM.

But in our scenario, in predicting cyber attack rates we can use the GRU instead of LSTM in the deep learning network as there are not many dependencies among features of the time series data.

Optimizers in deep learning networks

Optimizers of deep learning networks are the algorithms that are used to change the attributes of the network such as weights and learning rates in order to minimize the loss of the network. A good network should have a minimum loss value. The main optimizer which is used in the BRNN-GRU is an Adam Optimizer.

Adaptive Moment Estimation (Adam) optimizer has an adaptive learning rate that learns during the training of the network. These types of optimizers are better to use than static optimizers because the models learn and increase the accuracy with a number of training cycles. At the same time, the loss values should be decreased and adaptive with the number of training cycles. Adam optimizer works on first and second order momentums. It stores the exponentially decaying average of past squared gradients and the exponentially decaying average of past gradients. Therefore the Adam optimizer is fast and converges very rapidly. Also, it rectifies vanishing learning rates and high variance (Doshi, 2019).

Although Adam optimizer is the suitable optimiser for our scenario, there are a lot more optimizers that can be used in different neural networks. A detailed comparison is available in (Doshi, 2019).

Hyperparameters

The variables which determine the structure of the neural network are called Hyperparameters. The model accuracy and the performances can be changed according to different hyper parameter values. Weight, activation function, epochs, learning rates and optimizers are some of the common hyperparameters defined in neural networks. Hyperparameter tuning can help to increase the performance, accuracy and reduce the error rates. Grid search, random search or Bayesian optimization are common methodologies to optimize hyperparameters in deep neural networks (LoshchilovHutter, 2016). More information about hyperparameter tuning can be found from (LoshchilovHutter, 2016), (ReimersGurevych, 2017) and (Tensorflow, 2021)

Proposed Algorithm

INPUT : Historical cyber attack rate time series data

OUTPUT : Predicted values with Mean Squared Error

PROCESS :

1. Split the dataset into training, validation and testing data
2. Create small data frames (j) for training dataset inorder to minimize overfitting of the model by passing start index, end index and no.of rows per file
3. **for** Each mini-batch/data frame (j) **do**
4. Make each data frame compatible to pass GRU with specific shape
5. Shuffle the indices to get new order of files in the next iteration
6. Return GRU compatible dataframes
7. **end for**
8. Create small data frames (i) for testing dataset inorder to minimize overfitting of the model by passing start index, end index and no.of rows per file
9. **for** Each mini-batch/data frame (i) **do**
10. Make each data frame compatible to pass GRU with specific shape
11. Return GRU compatible dataframes
12. **end for**
13. Create the model with Bi directional RNN with GRU, set dynamically changeable hyperparameters
14. $k \leftarrow 0$
15. **while** $k < b$ iteration **do**
16. **for** Each mini-batch from training sample set (j) **do**
17. Compute k by performing forward and backward propagation
18. Update model using the Adam optimizer
19. **end for**
20. $k \leftarrow k + 1$
21. **end while**
22. Predictions $\leftarrow 0$
23. **for** Each mini-batch in the testing samples set (i) **do**
24. Compute $\hat{y}$ by performing forward and backward propagation using the trained model
25. Predictions $\leftarrow \hat{y}$
26. **end for**
27. **return** predicted value, MSE

Algorithm 8 : Pseudocode for predicting cyber attack rates using BRNN-GRU model

Methodology

Current Deep Learning Approaches for Cyber Attack Rate Prediction

As discussed in the literature review, deep learning prediction models are more commonly used in finance, natural language processing and many more fields. But when I was doing the literature review I found that there is a lack of research done for cyber attack rate prediction using deep learning models. Not only for the attack rate prediction, the categorization and other related cyber attack predictions are still at the research level.

At present, there are only a few approaches available in deep neural network models to analysing time series data. Recurrent neural networks are one example. In future, there will be more network types available for time series analysis and will support much more features exhibited in the time series data.

One of the major frameworks called BRNN-LSTM model (Fang et al., 2019) is a prominent framework that uses deep learning models to accommodate statistical features exhibited in the cyber attack time series. Also, it combines statistical features with artificial intelligence theories to increase performance and its prediction accuracy. The BRNN-LSTM framework is developed in Python and runs on the CUDA platform(Fang, 2018). This BRNN-LSTM model will be used in this paper for comparison purposes.

This model uses most of the latest techniques in deep neural networks to build the prediction model. As there are not many approaches to work with time series data, the only thing we can do so is using the same approach of BRNN-LSTM and tuning it with different hyper parameters. The model development section includes the ways to improve the current BRNN-LSTM framework and explains the use of other hyperparameters to increase performance and accuracy.

Data acquisition

Most of the major frameworks we discussed in the literature review have used CAIDA real time network telescope dataset(CAIDA, 2021), which is currently restricted access for the public. Hence for the training and the evaluation of BRNN-GRU we use the Kyoto University's Honeypot dataset (http://www.takakura.com/Kyoto_data/). This dataset contains time series data from 01-11-2006 to 31-12-2015.

This dataset consists of 24 main statistical features. 14 of them were extracted from the raw traffic of the honeypot systems deployed in Kyoto university networks. These conventional features are as follows.

1. **Duration** - The length of the connection is seconds
2. **Service** - Connection type such as http,ssh etc.
3. **Source Bytes** - The number of data bytes sent by the source IP address
4. **Destination Bytes** - The number of data bytes sent by the destination IP address
5. **Count** - The number of connections whose source IP and destination ID addresses are same for past 2 seconds
6. **Same_srv_rate** - Percentage of connections to the same service in count feature
7. **Serror_rate** - Percentage of connections that have "SYN" errors in count feature
8. **Srv_serror_rate**
9. **Dst_host_count**
10. **Dst_host_srv_count**
11. **Dst_host_same_src_port_rate**
12. **Dst_host_serror_rate**
13. **Dst_host_srv_serror_rate**
14. **Flag** - The state of the connection : example- REJ, S0, S1, OTH etc.(SONG et al., 2021)

This data was collected from Nepenthes, a low interaction honeypot service deployed on Kyoto University networks. More details about the features and the data about the Kyoto dataset can be found from (SONG et al., 2021).

Data preprocessing

The dataset consists of daily network connection details in a .txt file per day from 01-11-2006 to 31-12-2015. Therefore this model does not need to convert the connections to flows as the other frameworks did. The CAIDA telescope datasets come as .pcap files and for the analysis, they need to convert to flows.

This dataset is already preprocessed and removed irrelevant details by the research project done by (Song et al., 2011). We assume that they followed (Claffy et al., 1995) methodologies to preprocess the honeypot data, as (Claffy et al., 1995) is considered as a major reference for preprocessing honeypot time series data. According to (SONG et al., 2021) they have used the Bro 2.4 tool (The Bro Project, 2016) to convert traffic data to session data. Bro is an open source and powerful network analysis tool that can be used for network security monitoring. It is capable of monitoring traffic in very high performance environments(Cloud, 2019).

Before applying honeypot data preprocessing mechanisms mentioned in (Claffy et al., 1995), the connections where the status does not end with **FIN** or **RST** flags and connection lifetime more than 300s are removed from the individual dataset files (Zhan et al., 2013).

After completing the preprocessing step, we had to create a separate file including DateTime and the attack rates. We used a simple python function to archive that. The final input file looks like the below example.

Before passing the input data to the model, the input file is further pre-processed to remove null values from the dataset.

Date	Time	Attacks
13/01/2013	17:24:00	4216

Table 4: Sample data instance in the final input file

Selection of Technologies

We used Google Colaboratory, which is a free online cloud based Jupyter (Python) notebook environment that provides features for creating and training machine learning and deep learning models using CPUs, GPUs and TPUs. It supports Tensorflow and many more python based machine learning libraries such as Keras. The documentation and the support for Google Colaboratory, tensorflow and Keras are immense. Due to the cloud based feature of the Google Colaboratory, the researchers do not need to use their local computational power and they are able to use TPUs with high processing powers for free.

Tensor processing Units - TPU

TPUs are hardware accelerators designed by Google to provide high computational power for deep learning tasks. TPUs are more powerful than GPUs and CPUs which are also used in deep learning and 3D rendering like high end tasks .

GPUs are organized around programmable cores, while TPUs use a classic vector processor with a dedicated matrix multiply unit and excel the matrix according to the relevant task. Hence TPU matrix is an adaptable one.

When the user requests Cloud TPU v2 in Google Colaboratory, the user will get a virtual machine that has a PCI attached TPU board. This TPU board has 4 dual-core TPU chips. Each TPU core has a Vector Processing Unit and 128 x128 Matrix multiple Units which can provide much more processing power than other hardware accelerators.

Tensorflow

Tensorflow is a very popular, end-to-end open source platform for Machine learning tasks. Tensorflow contains most of the machine learning libraries, tools and community support which are very helpful for machine learning researchers. Google Colaboratory has its inbuilt tensorflow support. Tensorflow can ,

- Efficiently execute low level tensor operations on CPU, GPU or TPU
- Able to calculate the gradient of arbitrary differentiable expressions
- Scalable to many devices and parallelize them
- Export programs to external devices such as IOT, embedded systems and servers(Team, 2021). Hence Tensorflow has cross platform support

Keras

Keras is a high level, python based, machine learning API working on top of the Tensorflow-2 platform. It provides essential abstractions and building blocks for developing machine learning and deep learning models and ships them with high interaction velocities(Team, 2021).

Due to the abstractions and easy to use APIs, the researchers do not want to spend much more attention and time to develop the layers from scratch. The researchers can build their models in 3 ways using Keras on Tensorflow2. They can use sequential models, functional APIs or model subclassing approaches.

Model Development

As discussed in the previous chapters, the model introduced in this paper - BRNN-GRU framework for cyber attack prediction is built on Tensorflow using the Keras library.

For evaluation purposes, we have built the basic RNN-LSTM model and BRNN-LSTM model as well. The original BRNN-LSTM which was introduced by (Fang, 2018) has been developed using the Python NumPy library. The source code for Algorithm 6 is available on Fang's github repository although there is no source code available for Algorithm 5.

(https://github.com/xingfang912/time-series-analysis/blob/master/time_series_rolling.py). We made some assumptions when transforming the designed model from NumPy to Keras. Also, some of the hyperparameters were hardcoded in the original codebase. It might be because of tuning the models to support the CAIDA dataset efficiently. Apart from the number of RNN layers, the other hardcoded hyperparameters mentioned in the github repository is used to develop the model on Keras. The assumptions are commented on the source code on the Google Colabrary project.

The RNN-LSTM model is a basic LSTM RNN sequential model which has an LSTM layer and default hyperparameters.

The BTNN-LSTM is taken as the basement for the new prediction model presented in this paper. Hence the new model consists of the following features.

- Contain sequential model including RNN and Bidirectional GRU layers
- Support major statistical features of the cyberattack rates. Example: Long Range dependence and Extreme values
- Used adaptive optimizers for the layers - Adam Optimizer
- Use Rectified Linear Units(ReLU) as activation functions and backpropagation techniques with GRU in order to minimize the vanishing gradient problems in RNN networks
- The loss function is defined as Mean Squared Error
- The hyperparameters can be changed
- Mini batches of the data frames have been used to increase the efficiency of the computation and the accuracy
- Swap the mini batches in different iterations. Thus the model trains in different data variations and sequences.
- Runs on TPU. But this model is able to run on GPU and CPU as well.
- Use dropout layers for high regularizations of the model

The pseudo code for the algorithm is defined in the proposed Algorithm section.

The developed model can be found from the following URL.

https://colab.research.google.com/drive/1HAbhZOYi7ePRDE9pKZXODRLgN_M6VWrk?usp=sharing

Model Evaluation

Model evaluation is done using the Mean Squared Error(MSE) calculated for the trained model with validation dataset. The model with less MSE is considered the best model.

These values can be changed according to the time series data streams we pass for the model and the availability of the features. The dataset we use here - The Kyoto dataset might not contain extreme values or LRD features. Hence the model can give different predictions and accuracy levels for a different dataset.

Fang and his team had compared the evaluation of BRNN-LSTM models with Gray model, FARIMA+GARCH models, Hidden Markov models and marked point process model which are the currently available research for predicting cyber attack rates. They concluded that the BRNN-LSTM framework has higher accuracy, performance and lower MSE values when compared to the other models. By keeping that in mind, here we use only the RNN-LSTM and BRNN-LSTM models for the comparison of the BRNN-GRU model. The MSE values for these models are as follows.

Model	MSE
Baseline (Tested with a trained dataframe)	0.428345914375
RNN-LSTM	0.40692530679349045
BRNN-LSTM	0.39681923587779341
BRNN-GRU	0.37891233569034562

Table 5 : Calculated MSE values for the BRNN-GRU network

According to the above outputs, the BRNN-GRU model can provide much more accurate predictions than the other models. Moreover, the BRNN-LSTM with Keras provides more accurate results than basic RNN-LSTM models.

In the BRNN-GRU model, we replaced most of the hyperparameters and LSTM layers with GRU layers. The reason for the low MSE can be either hyperparameter tuning or usage of GRUs. However, GRU is not very good for complex and large datasets with high dependencies. Research found that a GRU can provide much more accurate results with less computational power for small datasets. As we pass the large datasets as small chunks of data frames to the BRNN-GRU model, GRU should perform better than LSTM.

Conclusions

We proposed a BRNN-GRU framework for predicting cyberattack rates ahead of a given time prior to the attack. This framework can accommodate complex phenomena exhibited by cyber attack rate datasets, such as long-range dependence, highly nonlinearity and extreme values. This Framework is designed to train with less computational power, high accuracy, minimized loss and error values. This framework is analysed against Kyoto university's Honeypot dataset.

Moreover, the research attention for the cyber attack prediction using deep learning has not got much more attention like other industrial fields. Fewer models available in deep learning for time series analysis and less knowledge of statistical features available in cyberattack time series make the researchers pay less attention to the cyber attack predictions.

According to the literature review conducted, we have already proved that deep learning models for cyber attack rate predictions are faster, use less computational power and provide high prediction accuracy with minimal error values. Deep learning models can skip the feature selection process in the machine learning models. The deep learning models can learn and train by themselves. Hence it can identify the actual features of the dataset, can learn with different sequences and minimise human errors. Apart from that, with the currently available models in deep learning, only the RNN can provide time series analysis which is suitable for cyber attack rate prediction. Therefore the only enhancements we can do for the cyber attack prediction models is to keep RNN as the base for the model and tune it with the best hyperparameter values. Also, we can minimise the loss and optimize the model with different approaches like introducing mini batches, backpropagation and directly working with RNN cells etc.

Although most of the researchers use RNN for the time series analysis, there are few ongoing researchers for the time analysis using deep learning frameworks. One of the approaches is timeGAN(Yoon et al., 2019) which uses Generative Adversarial Networks for the implementation. Recently, most of the researchers pay much attention to

introducing time series predictions using Generative Adversarial Networks where two models are trained simultaneously by an adversarial process. However, according to (Alberto Carrillo Romero, 2018) findings on his time series analysis research, the best performers were the Shallow LSTM with an accuracy of 74.16% and the GAN 72.68 % and the Deep LSTM 62.85% followed by ARIMA 59.57%(Alberto Carrillo Romero, 2018).

Hybrid model development in deep learning for time series analysis is another type of ongoing research. CNN-LSTM is one of the examples where Convolution Neural Networks are combined with Recurrent Neural Networks.

Introducing the Attention mechanism for time series data analysis also conducted by different researchers. Attention mechanism provides accurate predictions also in presence of long input sequences according to (Del Pra, 2020) and (Rémy, 2021).

Facebook Prophet (TaylorLetham, 2017) is also significant research conducted recently, and it is an automatic forecasting procedure of time series data based on an additive model where non-linear trends are fit with yearly, weekly, and daily seasonality, plus holiday effects. It works best with time series that have strong seasonal effects and several seasons of historical data. Prophet is robust to missing data and shifts in the trend, and typically handles outliers well. Therefore, before using Facebook Prophet in the cyberattack time series analysis, we have to confirm whether this data has strong seasonal effects and seasonal trends. The statistical features we currently know about the cyber attack time series data are, they exhibit long range dependencies, non linearly, extreme values and do not follow a Poisson's distribution. Moreover, AR-Nets - A simple auto-regressive Neural Network for time-series is also introduced by (J. Triebe et al., 2019) which facilitates time series analysis inside the neural networks and will be useful in cyber-attack predictions. Also, one of the ongoing research projects is Neural Prophet (Triebe, 2020) which acts as a bridge between Time series and ML knowledge. This framework is based on the Facebook Prophet framework and AR-Net. I assume in

the future, the researchers will use these new research findings to improve cybersecurity predictions.

To conclude the paper, although many solutions have been introduced in the cybersecurity prediction, detection and forecasting fields, still, there is no definite model for predicting all of these with 100% accuracy. Also, the newly designed time series prediction models should be tested with cyberattack rates time series as they comprise different statistical features. As technology evolves, the complexity of cyberspace and attackers' patterns also evolve. Hence this research area will always emerge with new findings and will be open as an important research topic.

References

- Alberto Carrillo Romero, R. (2018) *Generative Adversarial Network for Stock Market price Prediction*. [Online]. Available at: https://cs230.stanford.edu/projects_fall_2019/reports/26259829.pdf [Accessed: 12 March 2021].
- Almotairi, S., Clark, A., Mohay, G. and Zimmermann, J. (2008) Characterization of Attackers' Activities in Honeypot Traffic Using Principal Component Analysis. *2008 IFIP International Conference on Network and Parallel Computing*, p.147-154. *IEEE* [Online]. Available at: doi:10.1109/npc.2008.82 [Accessed: 23 January 2021].
- Amidi, A. and Amidi, S. (2021) CS 230 - Recurrent Neural Networks Cheatsheet. *Stanford.edu*. [Online]. Available at: <https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-recurrent-neural-networks#overview> [Accessed: 28 March 2021].
- Amini, A. and Soleimany, A. (2020) MIT Deep Learning 6.S191. MIT Deep Learning 6.S191. [Online]. Available at: <http://introtodeeplearning.com/> [Accessed: 20 May 2021].
- Bauwens, L. and Giot, P. (2000) The Logarithmic ACD Model: An Application to the Bid-Ask Quote Process of Three NYSE Stocks. *Annales d'Économie et de Statistique*, (60), p.117. *JSTOR* [Online]. Available at: doi:10.2307/20076257 [Accessed:18 February 2021].
- Bauwens, L., Giot, P., Grammig, J. and Veredas, D. (2004) A Comparison of Financial Duration Models via Density Forecasts. *INTERNATIONAL JOURNAL OF FORECASTING*, 20 (4), p.589–609. [Online]. Available at: <http://hdl.handle.net/1854/LU-8649410> [Accessed: 9 April 2021].
- CAIDA(2021) Historical and Near-Real-Time UCSD Network Telescope Traffic Dataset. [Online]. Available at: https://www.caida.org/catalog/datasets/telescope-near-real-time_dataset/ [Accessed: 4 April 2021].
- Choulakian, V. and Stephens, M. (2001) Goodness-of-Fit Tests for the Generalized Pareto Distribution. *Technometrics*, 43 (4), p.478-484. [Accessed: 19 December 2020].
- Claffy, K., Braun, H. and Polyzos, G. (1995) A Parameterizable Methodology for Internet Traffic Flow Profiling. *IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATION*, 13 (8), p.1481-1494. [Accessed: 13 February 2021].
- Cloud, A. (2019) How to Install Bro IDS on Ubuntu 16.04. *Medium*. [Online]. Available at: <https://alibaba-cloud.medium.com/how-to-install-bro-ids-on-ubuntu-16-04-ce81d759ce1c> [Accessed: 22 November 2020].
- Cryer, J. and Kung-Sik, C. (2008) *Time Series Analysis With Applications in R*. Springer Texts in Statistics. [Accessed: 14 March 2021].
- Del Pra, M. (2020) Time Series Forecasting with Deep Learning and Attention Mechanism. *Medium*. [Online]. Available at:

<https://towardsdatascience.com/time-series-forecasting-with-deep-learning-and-attention-mechanism-2d001fc871fc> [Accessed:27 April 2021].

Doshi, S. (2019) Various Optimization Algorithms For Training Neural Network. *Medium*. [Online]. Available at: <https://towardsdatascience.com/optimizers-for-training-neural-network-59450d71caf6> [Accessed: 25 January 2021].

Fang, X. (2018) xingfang912/time-series-analysis. *GitHub*. [Online]. Available at: https://github.com/xingfang912/time-series-analysis/blob/master/time_series_rolling.py [Accessed: 7 March 2021].

Fang, X., Xu, M., Xu, S. and Zhao, P. (2019) A deep learning framework for predicting cyber attacks rates. *EURASIP Journal on Information Security*, 2019 (1), p.1-11. *Springer Science and Business Media LLC* [Online]. Available at: doi:10.1186/s13635-019-0090-6 [Accessed: 25 April 2021].

Görner, M. (2021) Keras and modern convnets, on TPUs | Google Codelabs. Google Codelabs. [Online]. Available at: <https://codelabs.developers.google.com/codelabs/keras-flowers-tpu#0> [Accessed:15 May 2021].

HERRERO, Á., ZURUTUZA, U. and CORCHADO, E. (2012) A NEURAL-VISUALIZATION IDS FOR HONEYNET DATA. *International Journal of Neural Systems*, 22 (02), p.1250005. *World Scientific Pub Co Pte Lt* [Online]. Available at: doi:10.1142/s0129065712500050 [Accessed: 12 January 2021].

Huang, C., Han, J., Zhang, X. and Liu, J. (2019) Automatic Identification of Honeypot Server Using Machine Learning Techniques. *Security and Communication Networks*, 2019, p.1-8. *Hindawi Limited* [Online]. Available at: doi:10.1155/2019/2627608 [Accessed: 31 March 2021].

Hunter, S., Irwin, B. and Stalmans, E. (2013) Real-time distributed malicious traffic monitoring for honeypots and network telescopes. *2013 Information Security for South Africa. IEEE* [Online]. Available at: doi:10.1109/issa.2013.6641050 [Accessed: 5 December 2020].

Husak, M., Komarkova, J., Bou-Harb, E. and Celeda, P. (2019) Survey of Attack Projection, Prediction, and Forecasting in Cyber Security. *IEEE Communications Surveys & Tutorials*, 21 (1), p.640-660. *Institute of Electrical and Electronics Engineers (IEEE)* [Online]. Available at: doi:10.1109/comst.2018.2871866 [Accessed: March 17 2021].

J. Shimeall, T. and M. Spring, J. (2014) *Introduction to Information Security*. *Elsevier* [Online]. Available at: doi:10.1016/c2011-0-00135-7 [Accessed: 6 December 2020].

J. Triebe, O., Laptev, N. and Rajagopal, R. (2019) *AR-Net: A simple Auto-Regressive Neural Network for time-series*. [Online]. Available at: <https://arxiv.org/abs/1911.12436> [Accessed: 30 January 2021].

Justin, L. (2020) 3 Steps to Time Series Forecasting: LSTM with TensorFlow Keras - Just into Data. Just into Data. [Online]. Available at: <https://www.justintodata.com/forecast-time-series-lstm-with-tensorflow-keras/> [Accessed: 1 May 2021].
Kotenko, I. and Chechulin, A. (2013) A Cyber Attack Modeling and Impact Assessment Framework. *In: 5th International Conference on Cyber Conflict. Tallinn: NATO CCD COE Publications..* [Accessed: 21 January 2021].

Li, Y., Shi, L. and Feng, H. (2019) A Game-Theoretic Analysis for Distributed Honeypots. *Future Internet*, 11 (3), p.65. *MDPI AG* [Online]. Available at: doi:10.3390/fi11030065 [Accessed: 30 December 2020].

Loshchilov, I. and Hutter, F. (2016) CMA-ES For Hyper parameter Optimisation Of Deep Neural Networks. *ICLR 2016*. [Accessed: 23 March 2021].

Luk, K. (2019) Downloading Datasets into Google Drive via Google Colab. Medium. [Online]. Available at: <https://towardsdatascience.com/downloading-datasets-into-google-drive-via-google-colab-bcb1b30b0166> [Accessed: 13 April 2021].

Lutscher, P., Weidmann, N., Roberts, M., Jonker, M., King, A. and Dainotti, A. (2019) At Home and Abroad: The Use of Denial-of-service Attacks during Elections in Nondemocratic Regimes. *Journal of Conflict Resolution*, 64 (2-3), p.373-401. *SAGE Publications* [Online]. Available at: doi:10.1177/0022002719861676 [Accessed: 18 May 2021].

Meharchandani, D. (2021) 10 Major Cyber Attacks Witnessed Globally in Q1 2021 - Security Boulevard. *Security Boulevard*. [Online]. Available at: <https://securityboulevard.com/2021/04/10-major-cyber-attacks-witnessed-globally-in-q1-2021/> [Accessed: 3 June 2021].

Milkovich, D. (2019) 15 Alarming Cyber Security Facts and Stats | Cybint. *Cybint*. [Online]. Available at: <https://www.cybintsolutions.com/cyber-security-facts-stats/> [Accessed: 4 March 2021].

Muncaster, P. (2020) Cyber-Attacks Up 37% Over Past Month as #COVID19 Bites. *Infosecurity Magazine*. [Online]. Available at: <https://www.infosecurity-magazine.com/news/cyberattacks-up-37-over-past-month/> [Accessed: 22 May 2021]

Peng, C., Hu, T., Xu, S. and Xu, M. (2017) Modeling and predicting extreme cyber attack rates via marked point processes. *JOURNAL OF APPLIED STATISTICS*, 44 (14), p.2534–2563. [Online]. Available at: doi:http://dx.doi.org/10.1080/02664763.2016.1257590 [Accessed: 3 February 2021].

Reimers, N. and Gurevych, I. (2017) *Optimal Hyperparameters for Deep LSTM-Networks for Sequence Labeling Tasks*. [Online]. Available at: <https://arxiv.org/pdf/1707.06799.pdf> [Accessed: 29 November 2020].

Rémy, P. (2021) philipperemy/keras-attention-mechanism. *GitHub*. [Online]. Available at: <https://github.com/philipperemy/keras-attention-mechanism> [Accessed: 7 May 2021].

Safaei Pour, M., Mangino, A., Friday, K., Rathbun, M., Bou-Harb, E., Iqbal, F., Samtani, S., Crichigno, J. and Ghani, N. (2020) On data-driven curation, learning, and analysis for inferring evolving internet-of-Things (IoT) botnets in the wild. *Computers & Security*, 91, p.101707. *Elsevier BV* [Online]. Available at: doi:10.1016/j.cose.2019.101707 [Accessed: 16 February 2021].

Senanayaka, L. (2020) *Survey on Cyber Attack Detection, Prediction and Forecasting methodologies*. [Online]. Available at: <https://docs.google.com/document/d/1j5aFTOgxu2fZC5IWYgIPDM53TUYxgS0Fm6PgKKwNGes/edit?usp=sharing> [Accessed: 30 May 2021].

Song, J., Takakura, H., Okabe, Y., Eto, M., Inoue, D. and Nakao, K. (2011) Statistical Analysis of Honeypot Data and Building of Kyoto 2006+ Dataset for NIDS Evaluation. *BADGERS '11 April 10-13, 2011, Salzburg.*, p.29-36. [Accessed: 20 May 2021].

SONG, J., Okabe, Y. and Takakura, H. (2021) *Description of Kyoto University Benchmark Data*. [Accessed: 29 May 2021].

Taylor, S. and Letham, B. (2017) *Forecasting at scale*. *PeerJ* [Online]. Available at: doi:10.7287/peerj.preprints.3190v2 [Accessed:23 February 2021].

Team, K. (2021) Keras documentation: About Keras. *Keras.io*. [Online]. Available at: <https://keras.io/about/> [Accessed: 16 April 2021].

Tensorflow, T. (2021) Introduction to the Keras Tuner | TensorFlow Core. *TensorFlow*. [Online]. Available at: https://www.tensorflow.org/tutorials/keras/keras_tuner [Accessed: 6 May 2021].

TensorFlow, T. (2021) The Sequential model | TensorFlow Core. TensorFlow. [Online]. Available at: https://www.tensorflow.org/guide/keras/sequential_model [Accessed: 27 April 2021].

TensorFlow, T. (2021) tf.keras.layers.GRU | TensorFlow Core v2.5.0. TensorFlow. [Online]. Available at: https://www.tensorflow.org/api_docs/python/tf/keras/layers/GRU [Accessed: 29 April 2021].

The Bro Project, T. (2016) *Bro Documentation Release 2.4.1*. [Online]. Available at: <http://www.ncsa.illinois.edu/People/jazoff/bro-2.4.1.pdf> [Accessed: 25 March 2021].

Torabi, S., Bou-Harb, E., Assi, C. and Debbabi, M. (2020) A Scalable Platform for Enabling the Forensic Investigation of Exploited IoT Devices and Their Generated Unsolicited Activities. *Forensic Science International: Digital Investigation*, 32, p.300922. *Elsevier BV* [Online]. Available at: doi:10.1016/j.fsidi.2020.300922 [Accessed: 5 May 2021].

Triebe, O. (2020) Neural Prophet Extendable and Scalable Forecasting. In: 40th International Symposium on Forecasting, 2020. [Online]. Available at: https://github.com/ourownstory/neural_prophet/blob/master/notes/Presented_at_International_Symposium_on_Forecasting.pdf [Accessed: 15 January 2021].

Vichaidis, N., Tsunoda, H. and Keeni, G. (2018) Analyzing darknet TCP traffic stability at different timescales. *2018 International Conference on Information Networking (ICOIN)*. *IEEE* [Online]. Available at: doi:10.1109/icoin.2018.8343098 [Accessed: 18 April 2021]

Yang, S., Yu, X. and Zhou, Y. (2020) LSTM and GRU Neural Network Performance Comparison Study: Taking Yelp Review Dataset as an Example. *2020 International Workshop on Electronic Communication and Artificial Intelligence (IWECAI)*. *IEEE* [Online]. Available at: doi:10.1109/iwecai50956.2020.00027 [Accessed: 3 January 2021].

Yoon, J., Jarrett, D. and van der Schaar, M. (2019) Time-series Generative Adversarial Networks. *33rd Conference on Neural Information Processing Systems (NeurIPS 2019)*,. [Accessed: 14 November 2020].

Zhai, Y. (2005) *TIME SERIES FORECASTING COMPETITION AMONG THREE SOPHISTICATED PARADIGMS*. [Accessed:6 February 2021].

Zhan, Z., Xu, M. and Xu, S. (2013) Characterizing Honeypot-Captured Cyber Attacks: Statistical Framework and Case Study. *IEEE Transactions on Information Forensics and Security*, 8 (11), p.1775-1789. *Institute of Electrical and Electronics Engineers (IEEE)* [Online]. Available at: doi:10.1109/tifs.2013.2279800 [Accessed: 26 February 2021].

Zhan, Z., Xu, M. and Xu, S. (2015) Predicting Cyber Attack Rates With Extreme Values. *IEEE Transactions on Information Forensics and Security*, 10 (8), p.1666-1677. *Institute of Electrical and Electronics Engineers (IEEE)* [Online]. Available at: doi:10.1109/tifs.2015.2422261 [Accessed: 28 February 2021].

Zhang, J. (2021) MA636: Introduction to stochastic processes. *Kent.ac.uk*. [Online]. Available at: <https://www.kent.ac.uk/smsas/personal/lb209/files/notes1.pdf> [Accessed: 22 April 2021].

Word Count

Word count excluding quotations, references, titles for figures and tables, acknowledgements and appendices is 11387.

Appendices

Appendix 1

	PMAD		PMAD'	
Period	FARIMA	ARMA	FARIMA	ARMA
1-hour ahead prediction ($h = 1, p = 0.5$)				
I	0.179	0.446	0.173	0.157
II	0.217	0.363	0.149	0.149
III	0.298	0.273	0.305	0.312
IV	0.548	0.526	0.126	0.106
V	0.517	0.529	0.424	0.411
5-hour ahead prediction ($h = 5, p = 0.5$)				
I	0.206	0.556	0.292	0.314
II	0.212	0.351	0.420	0.411
III	0.297	0.272	0.246	0.250
IV	0.847	0.838	0.226	0.207
V	0.526	0.555	0.414	0.417
10-hour ahead prediction ($h = 10, p = 0.5$)				
I	0.869	0.801	0.314	0.281
II	1.024	1.034	0.277	0.284
III	1.00	1.002	0.202	0.201
IV	0.648	0.627	0.282	0.490
V	0.982	0.952	0.402	0.412

Appendix 1: Prediction error analysis for the predictions fone by ARIMA and FARIMA models where $p = 0.5$ in Gray model